FLATTENING OVER THE POINTS OF A GIVEN SUBSET OF THE CLOSED FIBRE IS CUT OUT BY A CLOSED SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The scheme theoretic version of the local flattening statement. Same chapter flat.tex, label remark-flattening-local-scheme-theoretic, invoked at proof line 6950. Provides, for each point of the closed fibre, a closed subscheme of the base whose effect is exactly to force flatness at that point, so that the intersection over the given subset does the job.
- Limit preserving behaviour of the flatness functor at primes, and the finite presentation argument from the local flattening theorem. More on Algebra, label more-algebra-lemma-limit-preserving-flat-at-primes (proof line 6961); Same chapter flat.tex, the final part of the proof of theorem-flattening-local referred to at proof lines 6963-6965. The first shows the functor is limit preserving when the sources and the module are of finite presentation, and the second supplies the argument giving finite presentation of the closed subscheme.

Source reference. Original lemma 05LT: sources/stacks-chow/tags.txt, source lines 7247–7247. Navigation pointer for \ref{more-algebra-lemma-limit-preserving-flat-at-primes} used inside the counted span (line 6961); the argument that consumes it is retained in the span.

Source reference. Original remark 05PH: sources/stacks-flat/pinned-original-source.tex, source lines 6905–6905. Navigation pointer for \ref{remark-flattening-local-scheme-theoretic} used inside the counted span (line 6950); the argument that consumes it is retained in the span.

Source reference. Original theorem 05PG: sources/stacks-flat/pinned-original-source.tex, source lines 6790–6790. Navigation pointer for \ref{theorem-flattening-local} used

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

inside the counted span (line 6965); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be the spectrum of a henselian local ring with closed point s. Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $E \subset X_s$ be a subset. There exists a closed subscheme $Z \subset S$ with the following property: for any morphism of pointed schemes $(T, t) \to (S, s)$ the following are equivalent*

(1) *$\mathcal{F}_T$ is flat over T at all points of the fibre X_t which map to a point of $E \subset X_s$, and*
(2) *$\mathrm{Spec}(\mathcal{O}_{T,t}) \to S$ factors through Z.*

Moreover, if $X \to S$ is locally of finite presentation, $\mathcal{F}$ is of finite presentation, and $E \subset X_s$ is closed and quasi-compact, then $Z \to S$ is of finite presentation.

Proof. For $x \in X_s$ denote $Z_x \subset S$ the closed subscheme we found in Remark 05PH. Then it is clear that $Z = \bigcap_{x \in E} Z_x$ works!

To prove the final statement assume X locally of finite presentation, $\mathcal{F}$ of finite presentation and Z closed and quasi-compact. First, choose finitely many affine opens $W_j \subset X$ such that $E \subset \bigcup W_j$. It clearly suffices to prove the result for each morphism $W_j \to S$ with sheaf $\mathcal{F}|_{X_j}$ and closed subset $E \cap W_j$. Hence we may assume X is affine. In this case, More on Algebra, Lemma 05LT shows that the functor defined by (1) is "limit preserving". Hence we can show that $Z \to S$ is of finite presentation exactly as in the last part of the proof of Theorem 05PG. □